

RED SEAL EXAM PREP

313A Refrigeration & AC Mechanic

Free 50-Question Mock Exam

Practice for the Red Seal / Certificate of Qualification exam

Pass mark: 70% · Aligned to the published occupational standard topic weightings

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Mock Exam — answer all questions, then check the key

Q1. An EPR (Evaporator Pressure Regulator) valve is installed in the suction line of a multi-evaporator system. Its function is to:

- A) Maintain a minimum evaporator pressure setpoint
- B) Limit maximum evaporator pressure to prevent overloading the compressor
- C) Regulate head pressure at the condenser outlet
- D) Balance refrigerant flow between multiple evaporators

Q2. When recovering refrigerant from a system with a compressor that has burned out internally, what special procedure is required?

- A) Flush the system with nitrogen before recovering to dilute the acid
- B) Use dedicated recovery equipment for acid-contaminated refrigerant and send it for reclamation
- C) Standard recovery with a standard recovery machine — a burnt compressor does not affect the recovery procedure
- D) Vent the refrigerant to atmosphere — contaminated refrigerant cannot be safely recovered

Q3. What is the purpose of a suction line accumulator in a refrigeration system?

- A) To subcool liquid refrigerant before the metering device
- B) To store additional refrigerant charge for use during peak demand
- C) To catch liquid refrigerant and oil before they reach the compressor
- D) To separate oil from the discharge gas before it enters the condenser

Q4. A system is cooling poorly. Suction pressure is low and evaporator superheat is high, but subcooling at the condenser outlet is higher than normal, and the filter drier is noticeably colder at its outlet than at its inlet, with sweat forming on it. What does this point to?

- A) A restricted liquid line filter drier holding liquid back upstream
- B) A TXV stuck wide open and overfeeding refrigerant into the evaporator
- C) An undercharge, since low suction with high superheat means low refrigerant
- D) An overcharge, because high subcooling comes from too much refrigerant

Q5. A refrigeration contractor is servicing the rack system in a supermarket owned by a private grocery chain, on privately owned land in a Canadian city. Which halocarbon rules govern how the refrigerant is handled on that job?

- A) The Transportation of Dangerous Goods Regulations and the TC recovery cylinder specification
- B) The Federal Halocarbon Regulations, 2022, made under the Canadian Environmental Protection Act, 1999
- C) The federal Ozone-depleting Substances and Halocarbon Alternatives Regulations
- D) The halocarbon regulation of the province or territory where the work is done

Q6. R-410A operates at significantly higher pressures than R-22. What is the approximate high-side pressure of R-410A at 35°C condensing temperature?

- A) 380 psi
- B) 450 psi
- C) 290 psi
- D) 175 psi

Q7. Why must polyol ester (POE) oil be kept in a sealed container and its open time on site kept short?

- A) POE stops mixing with refrigerant once exposed to direct daylight
- B) POE thickens on contact with air and will no longer reach bearings
- C) POE evaporates once opened, so the container loses usable volume
- D) POE pulls moisture out of the air quickly and then holds onto it

Q8. A refrigeration system fitted with a thermostatic expansion valve is running with normal suction pressure, discharge pressure well above normal for the outdoor temperature, and compressor amp draw above normal. The MOST likely cause is:

- A) Worn or leaking compressor valves
- B) An undercharge of refrigerant
- C) Poor condenser heat rejection
- D) The expansion valve stuck closed

Q9. Why is a small flow of dry nitrogen kept moving through the tubing while joints are being brazed?

- A) It cools the joint from the inside so the braze alloy sets more quickly
- B) It proves the joint is leak free as soon as the alloy has solidified
- C) It stops copper oxide scale forming inside the tube as it is heated
- D) It carries the flux through the joint and spreads it around the fitting

Q10. A hot gas defrost system is used in commercial refrigeration. How does it work?

- A) Electric heating elements mounted on the evaporator fins are energized during defrost
- B) The refrigerant charge is partially recovered into a storage tank, allowing the evaporator to warm naturally
- C) Hot compressor discharge gas is routed through the evaporator to melt frost
- D) Hot ambient air is blown over the evaporator to melt frost

Q11. After repairing a leak on a commercial condensing unit, a technician pressure tests with dry nitrogen at the test pressure marked on the nameplate, holds the test overnight without loss, releases the nitrogen to atmosphere, closes the system and weighs in the nameplate charge. On start-up the unit cools poorly and repeatedly trips its high-pressure control. What is the most likely cause?

- A) Nitrogen left in at atmospheric pressure because the system was not evacuated
- B) Compressor valves damaged by holding the nitrogen at test pressure overnight
- C) Moisture that entered during the repair and is freezing at the metering device
- D) A filter drier that was not replaced and has plugged with debris from the repair

Q12. Under CSA B52, what primarily determines whether a refrigeration system must be isolated in a dedicated machinery room rather than installed in an occupied space?

- A) The refrigerant safety group and system charge relative to occupancy and room volume
- B) The compressor type and the number of compressors serving the refrigeration system
- C) The supply voltage and the ampacity of the disconnect serving the condensing unit
- D) The footprint of the condensing unit relative to the available mechanical room area

Q13. What is the function of a high-pressure cutout switch in a refrigeration/air conditioning system?

- A) To limit compressor speed when head pressure is elevated
- B) To bypass the TXV when discharge pressure is too high
- C) To open the relief valve on the refrigerant cylinder when pressure exceeds maximum
- D) To shut down the compressor when discharge pressure exceeds a safe limit

Q14. What type of oil is required for systems using HFC refrigerants (such as R-134a, R-404A, R-410A)?

- A) POE (Polyolester) oil — HFCs are not miscible with mineral oil
- B) Alkylbenzene oil — this is the universal oil compatible with all refrigerant types
- C) PAG (Polyalkylene Glycol) oil — this is standard for all refrigeration applications
- D) Mineral oil — the same oil used with R-12 and R-22 systems

Q15. What is the purpose of a check valve in a refrigeration system, and where is it typically installed?

- A) To prevent oil from leaving the compressor — installed at the compressor suction port
- B) To regulate refrigerant flow rate — installed in the liquid line as a primary metering device
- C) To prevent backflow during the off-cycle — installed in heat pump discharge lines
- D) To limit maximum system pressure — installed at the compressor discharge as a pressure relief

Q16. An air-to-air heat pump in heating mode has lost most of its capacity. The outdoor coil is packed solid with ice, the machine has run for hours and has never entered a defrost cycle, and suction pressure is very low. Where should the technician look first?

- A) The refrigerant charge, because ice on a coil always means low charge
- B) The auxiliary heat strips, which are not being staged on to help
- C) The defrost initiation sensor and the control board reading it
- D) The reversing valve, which must be stuck between heating and cooling

Q17. A reciprocating compressor has a broken valve reed. How would this affect compressor performance and what test would confirm it?

- A) Broken valve reed raises head pressure — confirmed by discharge pressure higher than normal
- B) Broken reed allows gas to bypass compression — confirmed by a very low compression ratio
- C) Broken valve reed causes the compressor to short-cycle — confirmed by rapid on/off cycling
- D) Broken valve reed only affects oil pressure — confirmed by low oil pressure switch tripping

Q18. An electronic expansion valve (EEV or EXV) offers advantages over a TXV because:

- A) It is completely self-contained and never needs calibration
- B) It sets a fixed opening at start-up and needs no superheat feedback
- C) It requires no electrical power and operates purely on refrigerant pressure
- D) It meters flow from superheat computed by its controller

Q19. The refrigerant R-600a (isobutane) is used in domestic refrigerators. What special precaution is required when servicing these units?

- A) It is a flammable A3 hydrocarbon — no ignition sources, rated recovery equipment
- B) R-600a requires POE oil, and beyond that oil change no precautions are needed
- C) R-600a runs at very high pressure, so ordinary service gauges cannot be connected
- D) R-600a has a very high GWP, so the charge must go into a high-pressure cylinder

Q20. The sight glass in a refrigeration liquid line shows bubbles continuously while the system runs normally. This indicates:

- A) The condenser is working too efficiently
- B) Insufficient subcooling causing flash gas
- C) Normal operation — bubbles are expected in the liquid line
- D) Air in the refrigerant

Q21. A scroll compressor is producing a loud metallic grinding noise immediately on startup that clears after a few seconds. Oil level in the sight glass is normal. What is the MOST likely cause?

- A) The compressor motor windings are shorting, causing vibration and noise on startup
- B) The compressor discharge valve is worn, allowing high-pressure gas to blow back
- C) The compressor is worn out — metallic noise on startup always indicates imminent failure
- D) Liquid refrigerant migrated to the compressor during the off cycle — slugging on startup

Q22. Warm humid air passes over an evaporator coil and water collects in the drain pan. Why does the coil take moisture out of the air?

- A) The refrigerant absorbs the water vapour and carries it outdoors
- B) The low pressure inside the coil draws water vapour from the airstream
- C) The drain pan draws moisture out of the air that passes above it
- D) The coil surface is below the dew point, so moisture condenses

Q23. A packaged rooftop unit (RTU) is running but not cooling adequately. Both refrigerant circuits on this dual-circuit unit show normal pressures. The space temperature is 72°F (22°C) while the cooling setpoint is 68°F (20°C). What should be checked?

- A) The thermostat — replace the thermostat if temperature is not being achieved
- B) The compressors — short-cycling on the low-pressure control cuts run time too short
- C) The economizer or fresh-air damper stuck open, adding outdoor air heat load
- D) The refrigerant charge — add refrigerant until the temperature setpoint is reached

Q24. A technician is called to a system with oil logging (oil trapped in the evaporator). What causes oil logging and how is it corrected?

- A) Oil logging occurs only in systems with TXV valves — replace with cap tube to correct
- B) Oil leaving the compressor is not returning — low suction line velocity or wrong oil type
- C) Oil logging is caused by excessive oil charge — drain excess oil from the system and recheck performance
- D) Oil logging is caused by water contamination mixing with the oil — replace filter drier and all oil

Q25. A technician checks the "condenser split" on an air-cooled condensing unit. The condensing saturation temperature is compared against which temperature?

- A) The liquid line temperature leaving the condenser outlet
- B) The return air temperature entering the evaporator coil
- C) The dry-bulb temperature of the air entering the condenser
- D) The wet-bulb temperature of the air leaving the condenser

Q26. A refrigeration system has higher-than-normal suction pressure AND higher-than-normal discharge pressure simultaneously. What does this pattern most likely indicate?

- A) Refrigerant overcharge or poor condenser heat rejection (dirty coil, failed fan, or high ambient)
- B) Refrigerant undercharge — low charge causes both pressures to be low
- C) A restricted liquid line — restriction increases discharge pressure and starves the evaporator
- D) Loss of compressor efficiency — worn valves make both pressures converge toward the middle

Q27. A refrigeration system compressor contactor has pitted, burned contacts. What is the MOST LIKELY cause?

- A) Heating of the closed contact faces by the compressor's normal running current
- B) The contactor coil voltage is too high, causing excessive current through the contacts
- C) The contactor is rated too large for the duty, and oversized contacts pit sooner
- D) Repeated arcing at the contacts caused by compressor inrush or short cycling

Q28. What is the purpose of the condenser in a refrigeration system?

- A) To compress refrigerant vapour for circulation
- B) To reject heat and condense the refrigerant to a liquid
- C) To evaporate the refrigerant using heat from the room
- D) To regulate refrigerant flow to the evaporator

Q29. Refrigerants R-717 (ammonia) and R-744 (carbon dioxide) both carry numbers in the 700 series. What does that series indicate?

- A) A hydrocarbon refrigerant, numbered by its number of carbons
- B) A zeotropic blend, numbered in the order it was registered
- C) An inorganic compound, numbered as 700 plus its molecular mass
- D) A high-pressure refrigerant, numbered by its working pressure

Q30. A medium-temperature refrigeration system is running on a call for cooling, with the box still well above setpoint, and the suction gauge reads below atmospheric pressure (a vacuum). This indicates:

- A) Discharge valves leaking inside the compressor
- B) The system has been overcharged with refrigerant
- C) A severe undercharge or a blocked metering device
- D) Pump-down mode with the solenoid valve closed

Q31. A walk-in cooler compressor trips repeatedly on the low-pressure cutout (LPC) during normal winter operation, yet the cooler is maintaining setpoint temperature. What is the MOST LIKELY cause and the correct remedy?

- A) Low ambient is dropping head pressure and starving the TXV — install a head pressure control
- B) The LPC setpoint is factory-set too high — lower the cut-out in the field to stop the nuisance trips
- C) A restricted liquid line filter drier is reducing refrigerant flow and causing low suction pressure — replace the drier
- D) Refrigerant overcharge is flooding the evaporator and lowering suction pressure — recover excess refrigerant

Q32. What is the purpose of a phase monitor relay in a three-phase refrigeration system?

- A) To monitor refrigerant phase changes and shut down the system if the refrigerant is undercharged
- B) To balance the electrical load across three phases to reduce energy consumption
- C) To protect three-phase motors from phase loss, phase reversal, and phase imbalance
- D) To convert single-phase power to three-phase for use with commercial compressors

Q33. Which component in a refrigeration system is responsible for the pressure drop that causes the refrigerant to change from high-pressure liquid to low-pressure liquid/vapor mixture?

- A) The filter drier
- B) The metering device
- C) The compressor
- D) The evaporator

Q34. What is the difference between refrigerant recovery, reclaim, and recycling?

- A) Recovery, reclaim, and recycling are all the same process with different names used by different manufacturers
- B) Recovery: venting refrigerant safely through a filter. Reclaim: any reuse of recovered refrigerant. Recycling: destroying old refrigerant
- C) Recovery: removing to a container. Recycling: basic cleanup for reuse. Reclaiming: reprocessing to virgin purity
- D) Recovery and reclaim are identical; recycling means mixing refrigerants for reuse

Q35. A split system air conditioner is not cooling and the outdoor unit is not running, but the indoor air handler fan is running. What should be checked first?

- A) Refrigerant charge level — low charge prevents outdoor unit from starting
- B) Power to the outdoor unit: breaker, disconnect, and contactor
- C) The evaporator coil for ice formation blocking airflow
- D) The TXV sensing bulb position

Q36. In the refrigeration cycle, what state (phase) is the refrigerant as it leaves the condenser under normal operating conditions?

- A) A saturated mixture of liquid and vapour at the condensing temperature
- B) Supercritical fluid, past the critical point and no longer condensable
- C) Subcooled liquid, cooled below its condensing saturation temperature
- D) Superheated vapour, still well above its own saturation temperature

Q37. Why must a recovery cylinder never be filled with liquid refrigerant to the top of its internal volume?

- A) The recovery machine cannot pull a vacuum on a cylinder holding liquid
- B) A blend separates into its components once the vapour space is gone
- C) The cylinder valve passes vapour only and will seize on liquid flow
- D) Trapped liquid has no room to expand when warmed and can burst it

Q38. A single-phase capacitor-start hermetic compressor hums, draws locked-rotor current and trips its overload. Supply voltage at the compressor terminals is correct. What should be checked next?

- A) The refrigerant charge, since low charge stops a motor starting
- B) The condenser fan blade pitch, since a bent blade loads the motor
- C) The thermostat setpoint and the indoor blower speed setting
- D) The start capacitor and the start relay in the start circuit

Q39. Why has R-22 refrigerant been phased out in Canada?

- A) It is too expensive to produce
- B) It is not efficient enough for modern systems
- C) It reacts with copper tubing
- D) It is an ozone-depleting HCFC

Q40. A low-pressure control (LPC) is set to cut out at 20 PSI and cut in at 45 PSI on an R-22 system. The LPC trips repeatedly. The MOST likely cause is:

- A) Condenser fan failure causing high head pressure
- B) Compressor valves leaking
- C) Thermostat setting too cold
- D) Low charge causing suction pressure to fall

Q41. After a compressor burnout, what contamination must be addressed before installing a replacement compressor, and what is the procedure?

- A) Acid, carbon, and moisture must be removed: flush, burnout driers, oil acid check, deep evacuation
- B) Only the refrigerant oil needs replacement — flush the system with new refrigerant twice and recharge
- C) Install the new compressor and run it until the contamination is naturally removed through normal operation over several weeks
- D) Only the condenser and evaporator need flushing — the compressor discharge and suction lines self-clean during operation

Q42. Why can a small mass of refrigerant absorb so much heat in the evaporator?

- A) Because the compressor pushes it through the coil faster than a liquid
- B) Because a change of state absorbs far more heat than a temperature rise
- C) Because the pressure drop at the metering device chills it below the air
- D) Because its temperature rises steadily from the coil inlet to the outlet

Q43. A commercial refrigeration system uses a floating head pressure control strategy. What does this mean and what is its benefit?

- A) The discharge pressure is fixed at the minimum possible setpoint regardless of ambient temperature
- B) Head pressure fluctuates freely without controls, reducing wear on condenser fan motors
- C) Condenser fans are controlled to let head pressure drop as low as ambient allows
- D) The head pressure is allowed to rise and fall with refrigerant charge level, reducing the need for precise charging

Q44. A liquid receiver on a supermarket rack is fitted with a pressure-relief valve. Under CSA B52, the relief discharge must be:

- A) Vented into the machinery room at low level so the vapour can be exhausted
- B) Piped back into the suction line so the refrigerant is not lost
- C) Left open at the valve outlet so that a discharge can be seen and heard
- D) Piped to the outdoors, clear of building openings and air intakes

Q45. A hermetic compressor has a locked rotor condition. The FIRST check before replacement is:

- A) Replace the compressor immediately — locked rotor is always a mechanical failure
- B) Recharge the system with refrigerant
- C) Check for liquid flood-back and verify supply voltage
- D) Check the thermostat setpoint

Q46. A technician needs a cylinder to recover refrigerant from a system in Canada. Which cylinder is acceptable?

- A) A shop-built vessel fitted with a relief valve and a pressure gauge
- B) A cylinder marked to a Transport Canada specification and in date
- C) Any cylinder rated above the refrigerant's saturation pressure
- D) The disposable cylinder the new refrigerant charge was supplied in

Q47. A single-phase rooftop condensing unit's compressor hums and trips its overload on every start attempt. During a start attempt, voltage measured at the outdoor disconnect holds at 240 V while voltage measured at the compressor terminals collapses to about 190 V. What does the difference between those two readings localize?

- A) A supply problem at the utility transformer feeding the building
- B) A failed start capacitor unable to give the motor its starting torque
- C) High resistance between the disconnect and the compressor terminals
- D) A seized compressor drawing locked rotor current until it overheats

Q48. When recovering refrigerant from a system, the technician must:

- A) Release it slowly to prevent pressure spike in the room
- B) Mix it with the new refrigerant cylinder to save cost
- C) Release it outdoors if it is a small quantity (under 1 kg)
- D) Use a certified recovery machine and approved cylinder

Q49. A split-system air conditioner produces adequate cooling but the indoor unit is producing water dripping from the front panel into the conditioned space rather than draining properly. What is the MOST common cause?

- A) The condensate drain is blocked or the drain pan slope is insufficient
- B) The refrigerant charge is too high, causing condensation on the exterior of the refrigerant lines
- C) The indoor fan speed is too high, blowing condensate off the evaporator coil
- D) Low refrigerant charge is causing excessive coil icing, which melts and overflows

Q50. A walk-in freezer evaporator on electric defrost initiates every 6 hours and terminates either on its termination thermostat, set for this coil at 10°C (50°F), or on a 30-minute time limit. Every cycle runs the full 30 minutes and ends on the timer, yet when the technician opens the box at the end of a cycle the coil is completely clear of frost and the drain pan is hot. This indicates:

- A) Normal operation — defrost cycles run a fixed length regardless of need
- B) The defrost heaters are undersized for the coil and its load
- C) The termination thermostat is faulty or positioned incorrectly
- D) Refrigerant undercharge is causing excess frost accumulation

Answer Key & Explanations

Q1. Answer: A

EPR valve: sets minimum evaporator pressure (warmer setpoint), preventing the evaporator from getting too cold. In multi-temperature systems, different evaporators need different temperatures. The EPR valve limits how cold a warmer evaporator can get by maintaining minimum suction pressure above the lower-temperature coil setpoint. Example: produce case at 35°F uses an EPR; freezer evaporator at -10°F does not.

Q2. Answer: B

Burned-out compressor: refrigerant is acid-contaminated — use dedicated equipment and send for reclamation. A burned motor creates acid and carbon deposits from the breakdown of motor windings (copper, varnish, refrigerant oil). This acid-contaminated refrigerant cannot be recycled back into a system — it must be sent to a reclamation facility (not just recycling). Use dedicated "burnout" recovery equipment to prevent contaminating your regular recovery machine, and clearly label the recovery cylinder as contaminated.

Q3. Answer: C

Suction accumulator: liquid trap in suction line that holds back liquid refrigerant and oil that might otherwise enter and damage the compressor (liquid slugging). Positioned between the evaporator and compressor, it allows vapor to pass through while holding back liquid. A small metered orifice slowly returns trapped oil back to the compressor. Common on heat pumps (liquid can return during defrost) and systems with large refrigerant charges. Oil trapped too long = compressor oil starvation.

Q4. Answer: A

A restriction in the liquid line starves the coil in exactly the way an undercharge does, and subcooling is what separates the two. Low suction pressure with high superheat says one thing only: the evaporator is not getting enough refrigerant. Both faults produce it, so look upstream to tell them apart. An undercharge leaves too little liquid in the condenser, so subcooling reads LOW. A restriction leaves the compressor backing liquid up behind it, so subcooling reads HIGH while the coil downstream goes hungry.

The temperature drop across the drier is the direct evidence. Refrigerant is expanding through it as though it were a second metering device, losing pressure and therefore losing saturation temperature, which is why the outlet is cold enough to sweat — on a badly plugged drier, cold enough to frost — while the inlet stays warm. Recover, replace the drier, and treat the restriction as evidence rather than simply as a failed part: a drier that plugs and then plugs again says the system is dirty, so look for moisture, acid or burnout residue and check the metering device screen while you are in there. The remaining answers do not fit the readings. An overcharge does raise subcooling, but it raises suction pressure and lowers superheat rather than starving the coil, and it puts no temperature drop across the drier. A valve stuck wide open floods the coil, giving low superheat with high suction pressure — the opposite of what is measured.

Q5. Answer: D

On a privately owned commercial system on private land, the halocarbon rules that bind the technician are the province's or territory's own, not the federal ones. The Federal Halocarbon Regulations, 2022 (SOR/2022-110), made under the Canadian Environmental Protection Act, 1999, open with an application section: they apply to air-conditioning or refrigeration systems, solvent systems, fire-extinguishing systems and containers located in Canada that are owned by Her Majesty in right of Canada, a board or agency of the Government of Canada, a Crown corporation as defined in the Financial Administration Act, or a federal work or undertaking, or that are located on aboriginal lands or federal lands. A supermarket owned by a private grocery chain on private land is none of those, so that instrument does not reach this job.

Inside its own scope the federal regulation is strict, and it is worth knowing in its own terms because federal sites do come up. It provides that a person must not release, or allow or cause the release of, a halocarbon contained in an air-conditioning or refrigeration system, a fire-extinguishing system or a container, or in equipment used in the reuse, recycling, reclamation or storage of a halocarbon. The prohibition does not apply if the release is for the purpose of calibrating leak-detecting devices with equipment designed specifically for that purpose and the manufacturer's recommended procedures are followed; results from connecting or disconnecting hoses that are less than 1 m in length and used for charging with, or recovering, a halocarbon; results from a purge system on an air-conditioning or refrigeration system, including any associated recovery equipment, that emits less than 0.1 kg of halocarbon per kilogram of air purged; or is from a fire-extinguishing system for the purpose of fighting a fire that is not set for training purposes, or of testing the system in a military vehicle as authorized by a permit. Note the boundaries as written: a hose of exactly one metre is not inside the hose exception, and a leak-detector calibration is only inside its exception when both conditions, the purpose-built equipment and the manufacturer's procedure, are met. The regulation also provides that only a certified person may install, service or recover a halocarbon from an air-conditioning system or refrigeration system. None of that changes who the regulation applies to.

Outside that scope, Environment and Climate Change Canada describes provincial and territorial legislation on ozone-depleting substances and halocarbon alternatives as complementing the federal regulations, and lists among its requirements proper labelling and handling of equipment, training for equipment service providers, procedures for installing, removing, servicing, repairing or decommissioning products, and a prohibition on recharging leaking equipment. So the rules that actually bind this job, meaning who may handle the refrigerant, what has to happen to it instead of a release, and what has to be recorded, come from the halocarbon regulation of the province or territory where the work is done, and they differ in detail from one province to the next.

The other two instruments are federal and are real, which is what makes them tempting. The Ozone-depleting Substances and Halocarbon Alternatives Regulations control the manufacture, import, export, sale and use of the substances themselves, through prohibitions, permits and allowances; they contain no release prohibition for a refrigeration system and say nothing about who may service one. The Transportation of Dangerous Goods Regulations do bear on this trade, but on the recovery cylinder, its TC specification and its movement on the road, not on how the system is serviced.

Q6. Answer: C

R-410A at a 35°C (95°F) condensing temperature sits near 295 psig on the high side, so 290 psi is the closest choice. R-22 at that same 35°C sits near 182 psig, so the gap at condensing conditions is about 60 per cent. Lower down the curve the gap is wider: at a 40°F evaporator temperature R-22 is near 69 psig while R-410A is near 118 psig, about 70 per cent higher. There is no single multiplier that covers the whole curve, which is why any comparison of the two has to name the saturation temperature it is made at, and has to compare gauge with gauge — reading one refrigerant as gauge pressure and the other as absolute makes the gap look far smaller than it is. On a hot day the measured head pressure runs higher still, because the condensing temperature sits well above ambient. All of this is why R-410A systems require components, hoses, gauges and recovery equipment rated for the higher pressure.

Q7. Answer: D

POE is hygroscopic: it absorbs atmospheric moisture far faster than mineral oil, and it holds that water chemically rather than simply carrying it along. That is what makes it a field problem. Free water in a system can be boiled off with a deep vacuum, but moisture bound into POE largely will not come out on the vacuum pump, so an open pail left on the truck for an afternoon can carry water straight into a circuit that was correctly evacuated. Water plus POE plus heat gives hydrolysis, hydrolysis gives acid, and acid gives copper plating, varnished valves and eventually a burnout. Working practice follows from that: buy oil in the smallest sensible container, keep the cap on until the moment of use, pour rather than leave it standing open, never return poured oil to the container, and fit a fresh liquid line drier whenever the circuit has been opened. POE is used in the first place because HFC and HFO refrigerants are not miscible with mineral oil, so this handling discipline is the price of that compatibility.

Q8. Answer: C

Head pressure that has climbed while suction pressure stayed where it belongs says the condenser is not getting rid of its heat. A condenser can only reject heat into what it is given, so a fouled or blocked coil, a fan that has failed or is turning the wrong way, discharge air recirculating around the unit, a high ambient, or non-condensable gas occupying condenser volume all force the refrigerant to condense at a higher temperature than it should, and the pressure rises with that temperature. The compressor is then pumping against a wider difference, doing more work, which is exactly what the raised amp reading is reporting.

Suction pressure stays near normal on this machine because the thermostatic expansion valve throttles to hold superheat and goes on feeding the coil at the rate the load calls for. The same fault reads differently on a fixed-orifice machine: there the higher liquid pressure pushes more refrigerant through the orifice, so suction pressure rises along with the head and the pattern becomes high on both sides.

The other three faults move the pressures in other directions. An expansion valve stuck closed starves the coil, so suction pressure falls well below normal and discharge falls with it. Worn or leaking compressor valves let gas bleed back instead of being pumped, so the machine still turns but pumps poorly: suction pressure runs higher than it should because the compressor cannot pull it down, discharge pressure runs lower than normal because it cannot build head, the two pressures come closer together, and amp draw falls rather than rises. An undercharge leaves too little refrigerant in the system, so suction pressure, discharge pressure and subcooling all read low together.

Q9. Answer: C

Copper heated in air grows a black, flaky oxide on the inside of the tube, and a slow trickle of nitrogen displaces the air so that scale never forms.

The scale does not stay where it was made. Refrigerant and oil wash it downstream, where it plugs the metering device screen, blinds the filter drier, scores bearings and ends up in the compressor — one of the classic reasons a system never runs right after a repair, and a fault that is invisible from outside the tube. The flow has to be small: enough to sweep the air out, not enough to pressurize the joint and blow the molten alloy out of it. Regulate it down, vent it at an open end so it can actually flow through, and keep it going until the joint has cooled.

Nitrogen carries away very little heat at that flow, and quenching a brazed joint is the last thing wanted anyway. Brazing copper to copper with a phosphorus-bearing alloy needs no flux inside the tube at all, and flux is never carried by a gas. Leak testing is a separate step, done after the joint has cooled, with nitrogen brought up to the specified test pressure. And never apply a flame to a system that still holds refrigerant: refrigerant meeting a flame or a hot surface breaks down into toxic products.

Q10. Answer: C

Hot gas defrost: discharge gas heats evaporator from inside — efficient and fast. During hot gas defrost, solenoid valves redirect superheated compressor discharge gas directly to the evaporator inlet (in some designs the coil is fed in reverse flow). The hot gas condenses inside the evaporator coil, releasing its heat of condensation directly to the frost accumulation. This is faster and more energy-efficient than electric defrost. Common in commercial walk-in coolers and display cases.

Q11. Answer: A

Letting the nitrogen out is not the same as taking it out. When the gauges read zero the system is still full of nitrogen at atmospheric pressure, and only a vacuum pump removes it.

The sequence after a nitrogen test is release, evacuate, then charge; this technician went from release straight to charging, so the refrigerant was weighed in on top of a system full of nitrogen. Nitrogen does not condense at any pressure or temperature the condenser can reach, so it is carried round to the condenser and stays there, occupying surface that should be rejecting heat and adding its own partial pressure to the refrigerant's. Head pressure climbs beyond anything the ambient and the load can justify, the high-pressure control trips, the unit restarts when the pressure falls back and trips again, and the little running it does is at reduced capacity. There is no bleeding it off: recover the charge, evacuate to a deep vacuum proven with a micron gauge on a standing test, and weigh the charge back in.

The other faults are real but each writes a different signature. A plugged filter drier is a liquid line restriction: it starves the evaporator, so suction pressure runs low with high superheat and the drier outlet runs colder than its inlet; head pressure does not climb and the high-pressure control has no reason to trip. Moisture freezing at the metering device also starves the coil, intermittently, with the same low-side picture and a moisture indicator turning colour; again nothing pushes the high side up. Compressor valve damage is the opposite of what was found: a compressor with damaged valves still turns and moves some gas but pumps poorly, so suction pressure runs higher than it should, discharge pressure lower, the two converge and amp draw falls, which is a machine that cannot reach its high-pressure cut-out rather than one tripping it repeatedly.

Q12. Answer: A

CSA B52 bases the requirement on the refrigerant's safety classification, the charge quantity, the occupancy, and the volume of the space. Each refrigerant has a Refrigerant Concentration Limit. If the full system charge released into the occupied space could exceed that allowable concentration, the system has to be isolated in a machinery room with refrigerant detection and mechanical ventilation. Higher-risk occupancies — institutional and public assembly, where people cannot readily leave — carry lower allowable limits than an ordinary commercial space. How the unit is arranged and powered does not enter into it: compressor type and count, the electrical supply, and how much floor the equipment takes up are installation questions, not the trigger for the machinery room.

Q13. Answer: D

High-pressure cutout: safety device that shuts down compressor at excessive head pressure. High head pressure can be caused by: dirty/blocked condenser, condenser fan failure, refrigerant overcharge, non-condensables in the system, or high ambient temperature. If head pressure reaches the cutout setpoint (typically 590–650 PSI for R-410A systems), the compressor is de-energized — protecting the compressor, condenser, and refrigerant circuit from over-pressure damage. The manual reset type requires a technician reset, preventing restart until the cause is investigated.

Q14. Answer: A

HFC refrigerants require synthetic POE (polyolester) oil for proper lubrication and oil return — mineral oil is not miscible and will cause oil logging and lubrication failure. R-12 and R-22 (CFC/HCFC) systems use mineral oil or alkylbenzene oil. HFCs (R-134a, R-404A, R-410A, R-407C) require POE oil, which mixes with HFCs so the oil is carried around the circuit and returned to the compressor. Mixing mineral oil with HFCs causes the oil to separate out and pool in the evaporator, starving the compressor. POE oil is strongly hygroscopic — it takes up atmospheric moisture quickly and holds it, so containers stay sealed and open time is kept to a minimum.

Q15. Answer: C

Check valve: allows flow in one direction only — prevents refrigerant flowing backward during the off-cycle. Used in discharge lines on heat pumps and tandem/parallel compressor systems. On heat pumps: discharge line check valves direct refrigerant correctly when the reversing valve switches modes. On parallel/tandem compressor systems: individual discharge line check valves prevent the running compressor from pushing hot discharge gas into the stopped compressor. Also used in liquid lines of multi-evaporator systems to prevent migration.

Q16. Answer: C

Frost on the outdoor coil is normal in heating — in that mode the outdoor coil is the evaporator and it runs below freezing and below the dew point of the outdoor air. What is abnormal is that the control never acted on it. A heat pump permits defrost either on time and temperature (a timer that runs only while the compressor runs, gated by an outdoor coil sensor showing the coil is cold enough to be frosting) or on demand, from the spread between coil temperature and outdoor ambient or from the pressure drop across the coil. If that coil sensor is open, shorted, drifted or clamped where it does not read coil temperature, the board never sees the condition that allows a defrost, and the machine ices itself solid while every other reading simply follows from the blocked coil. Check the sensor's resistance against the manufacturer's resistance-temperature table, check where and how it is clamped, and force a defrost at the board to prove the reversing valve shifts and the outdoor fan stops.

Low suction pressure and lost capacity here are consequences of an ice-blocked evaporator, not evidence of an undercharge; the charge cannot be judged at all until the coil has been thawed and the machine run clean. A reversing valve stuck part-way would show poor capacity in both modes with the pressures converging, not a defrost that never begins. Staging the auxiliary heat would hide the complaint while the outdoor coil stayed frozen.

Q17. Answer: B

Broken compressor valve reed: gas bypasses the damaged valve instead of being compressed, severely reducing compression efficiency and capacity. A broken suction reed lets cylinder gas escape back into the suction line on the compression stroke — suction pressure runs high because the compressor can no longer pull it down, discharge pressure runs low, and mass flow drops. A broken discharge reed lets compressed gas leak back into the cylinder on the suction stroke and re-expand — again suction pressure rises and discharge pressure falls. Either way the two pressures converge, so a fault that pushes head pressure up is not this signature. Amp draw is below normal because the compressor is doing less work, while discharge temperature climbs because the same gas is compressed over and over. Confirm by comparing the measured compression ratio against the ratio the system's design evaporating and condensing conditions should produce — there is no single healthy number, since a low-temperature system normally runs a far higher ratio than a comfort-cooling system. A valved-out pump-down check (front-seat the suction service valve and watch whether the compressor pulls the low side down and holds it) separates a worn compressor from a system fault. A reed failure does not by itself cycle the compressor on and off, and it is not confined to the oil circuit — it shows in the refrigerant-side pressures long before an oil pressure switch reacts. Verify with valve plate inspection before condemning the compressor.

Q18. Answer: D

EEV: electronic precision control based on multiple sensor inputs. The controller reads suction pressure from a transducer and suction line temperature from a sensor, calculates superheat from the two, and drives a stepper motor to reposition the valve — much faster response and a far wider control range than a mechanical TXV, which is why EEVs suit inverter-driven and VRF equipment. The other answers describe things an EEV is not. It is not a pressure-operated valve: without control power and a working controller it does not meter at all. It is not self-contained either — the valve is only one part of a package that includes the controller and its sensors, and the superheat target is a programmed setting that can be checked and adjusted. And it does not park at a fixed opening: superheat feedback is recalculated continuously, which is precisely the advantage being asked about.

Q19. Answer: A

R-600a (isobutane) is a highly flammable A3 refrigerant — flammability is the primary hazard. Hydrocarbon refrigerants (R-600a, R-290/propane) have excellent thermodynamic properties but they burn. They are used in small-charge domestic appliances, where the charge is small enough that a full leak stays below the lower flammability limit in the room. Servicing: ventilated areas only, no open flames or sparks, non-sparking tools, and recovery equipment rated for flammable gases. These appliances must not be serviced in an enclosed space. R-600a is a low-GWP refrigerant and it operates at lower pressures than the common HFCs, so neither a GWP argument nor a high-pressure argument is what drives the precautions.

Q20. Answer: B

Bubbles in sight glass = flash gas = subcooling problem. Liquid refrigerant should arrive at the metering device as a solid liquid (no bubbles). Bubbles (flash gas) indicate: refrigerant is partially vaporizing before the metering device — caused by low charge (low subcooling), restriction in liquid line (pressure drop = boiling), or high liquid line temperature.

Q21. Answer: D

Liquid refrigerant migration to compressor during off cycle causes liquid slugging on startup — grinding/rattling noise that clears as liquid is pumped out. The migrated liquid is often an oil-refrigerant mixture in the crankcase. Scroll compressors are less tolerant of liquid than reciprocating compressors — the scrolls momentarily separate but can still be damaged. Causes: low ambient temperature with refrigerant migrating to coldest point (compressor), crankcase heater failed/not installed, refrigerant overcharge. Prevention: crankcase heater, pump-down cycle on shutdown, check valve in suction line.

Q22. Answer: D

A cooling coil dehumidifies because its surface sits below the dew point of the air passing over it. Air holds water vapour; cool any surface below the temperature at which that vapour would begin to condense — the dew point — and water comes out of the air onto the surface, exactly as it does on a cold glass. On a coil it runs down the fins into the drain pan and leaves the building as condensate. Removing that moisture is latent cooling; lowering the air temperature is sensible cooling, and a coil does both at once, which is why a coil handling humid air spends part of its capacity on water rather than on temperature.

The refrigerant never touches the air — it is sealed inside the tube, so it can neither absorb the moisture nor be affected by it, and the low pressure inside that sealed circuit has no reach into the airstream. The drain pan only catches what has already condensed. Two practical consequences follow from the real mechanism: a coil running warmer than the dew point cools without dehumidifying, and a coil driven below freezing turns the condensate to ice and blocks its own airflow.

Q23. Answer: C

An RTU that will not hold setpoint while its refrigerant pressures read normal is telling you to look beyond the refrigerant circuit. Normal pressures mean the refrigerant system is working; it simply cannot overcome the load it is being handed. Investigate: an economizer damper stuck open or a fresh-air damper out of adjustment, either of which drags outdoor heat into the return; a dirty condenser coil; a dirty evaporator coil restricting airflow; duct leakage; missing supply duct insulation; return air bypassing the space; or a building load beyond the equipment's capacity on an extreme heat day. Normal pressures on both circuits also rule out a low-pressure-control cutout, and adding refrigerant to a system that already reads normal only overcharges it.

Q24. Answer: B

Oil logging: oil circulates out of the compressor but cannot return — inadequate oil circulation lets it accumulate in low points and the evaporator. Refrigerant velocity carries oil through the system. If suction line velocity is too low (system running at partial load, oversized or incorrectly sized/routed piping), oil drops out of the refrigerant stream and pools. Risers need oil traps (P-traps) sized to maintain velocity even at minimum load. Wrong oil type/viscosity: too thick to flow at evaporator temperature. Low refrigerant charge also reduces velocity. Correction: verify line sizing and re-pipe as needed, add traps/risers, correct refrigerant velocity, verify oil specification and compatibility.

Q25. Answer: C

Condenser split is the condensing saturation temperature minus the dry-bulb temperature of the air entering the condenser. An air-cooled condenser can only push its heat into the air it is given, so the entering air is the reference the condensing temperature has to be judged against. A condensing temperature on its own means nothing until you know whether the unit is sitting in cool morning air or in full afternoon sun. A split much wider than the unit was designed for says the coil cannot move its heat: fouled fins, a failed fan or one turning the wrong way, discharge air recirculating around the unit, blocked airflow, or non-condensables occupying condenser volume. A split narrower than design, on a unit that is cooling poorly, points the other way and suggests there is not enough refrigerant in the coil to load it. Note what the other comparisons actually give you: condensing saturation temperature against liquid line temperature is subcooling, a different measurement answering a different question, and wet-bulb belongs to evaporative equipment rather than a dry air-cooled coil.

Q26. Answer: A

High suction + high discharge simultaneously = refrigerant overcharge or a condenser that cannot reject heat. An overcharge backs liquid up in the condenser and raises both pressures; a dirty or blocked condenser, a failed condenser fan, or high ambient temperature prevents heat rejection with the same result. Check condenser cleanliness, fan operation, and refrigerant charge. Contrast this with worn compressor valves, where suction rises while discharge FALLS and the two pressures converge toward the middle — that is compressor inefficiency, not this pattern.

Q27. Answer: D

Pitted, burned contacts are arc erosion: the damage is done in the instants when the contacts make and break under load, and on a compressor circuit the routine high-energy event is the starting inrush. A compressor motor draws many times its running current at the moment it starts, before the rotor is up to speed. Every make against that current, every bounce of the contacts as they close, and every break under load draws an arc across the contact faces, and each arc carries a little metal away, so the faces roughen, pit and discolour. Short cycling multiplies the number of those arcs, which is why a contactor that keeps burning up is usually telling you the machine is starting far too often. Fitting another contactor without finding out why it is cycling, whether that is a control differential set too tight, a low charge cycling the low-pressure control, poor heat rejection cycling the high-pressure control, or equipment oversized for its load, only buys time.

Voltage belongs on the check list, and it is a real second cause rather than a distraction. A coil that cannot pull the armature fully in, because the control voltage is low, lets the contactor chatter, and a chattering contactor makes and breaks load current over and over and burns its contacts doing it. On packaged equipment the control transformer is fed from the line, so a low supply voltage lowers the coil voltage with it; measure both at the moment of a start, not with the machine idle. But the question asks for the most likely cause, and inrush arcing is present on every start of every compressor, while a chattering coil is a specific fault that also announces itself with noise. The three alternatives each rest on a misunderstanding. Steady running current in closed contacts does not pit them: a clean, fully seated contact carrying current within its rating runs cool, and the contact that overheats while closed is one that has already been pitted and now has a high-resistance face, so heating is a consequence of the damage rather than its cause. Coil voltage sits in its own circuit, isolated from the load contacts, so raising it overheats or burns the coil and does not increase the current the contacts carry. And a contactor with more contact mass than the duty demands interrupts the inrush more easily, so oversizing reduces contact erosion rather than causing it.

Failure signatures worth knowing: welded contacts mean the compressor will not shut off, open contacts mean no start, and high-resistance contacts show up as voltage drop across a closed pole under load and local overheating.

Q28. Answer: B

Condenser: rejects heat, converts vapour to liquid. High-temperature, high-pressure vapour from the compressor enters the condenser. Heat is rejected to the outside environment — ambient air (air-cooled) or water (water-cooled). As heat leaves, the vapour condenses into a high-pressure liquid ready for the metering device.

Q29. Answer: C

The 700 series is reserved for inorganic refrigerants, and the last two digits are the substance's molecular mass. Ammonia has a molecular mass of 17, so it is R-717; carbon dioxide is 44, so it is R-744; water is 18, so it is R-718; air is taken as 29, so it is R-729. The series is worth knowing because a number tells you immediately what family you are dealing with. The 400 series is zeotropic blends and the 500 series azeotropic blends, and both are numbered in the order they were registered rather than by what is in them — which is why R-407A and R-407C hold the same components in different proportions, are different refrigerants, and are not interchangeable. The 600 series covers organic compounds, including R-600a, isobutane. Halocarbons take their numbers from their chemical formula.

What a refrigerant number never encodes is a pressure rating or a safety property. Toxicity and flammability come from the separate ASHRAE safety group — the letter-and-number code such as A1, A2L or A3 — and pressure comes from the refrigerant's own saturation curve, which is why the pressure-temperature chart exists.

Q30. Answer: C

A vacuum on the low side of a running system means almost nothing is reaching the evaporator. With the box calling for cooling, the compressor is pulling vapour out of the coil faster than the metering device can replace it, so the pressure falls until it drops below atmospheric. Two faults do that: a severe undercharge, where there is barely any refrigerant left to feed the coil, and a blockage that stops the flow - a metering device stuck closed, a plugged strainer or filter-drier, or a kinked liquid line. Either way the compressor loses the vapour flow that returns its oil and, on a suction-cooled motor, carries the winding heat away, so shut the system down and diagnose before it is damaged. Running below atmospheric pressure also draws air and moisture into the low side through any leak, fitting or seal, which puts non-condensables and acid into the system. The other readings do not fit. An overcharge raises suction pressure rather than lowering it. Leaking compressor discharge valves let high-side gas bleed back into the cylinder, which pushes suction pressure UP while head pressure falls toward it. And a pump-down draws the low side down only after the call for cooling has ended and the liquid-line solenoid has closed, not while the box is still calling.

Q31. Answer: A

In cold weather the air passing over the condenser is colder, the condenser rejects heat too easily, and condensing (head) pressure falls. A thermostatic expansion valve is driven by the pressure difference across it, so it needs a minimum condensing pressure to push its rated flow. When head pressure sags, flow through the valve drops, the evaporator is starved, suction pressure falls below the low-pressure cut-out, and the compressor trips — even though the box is still holding setpoint, because the trips are short and the load is light in winter. The remedy is a head pressure control that holds condensing pressure up as ambient falls: condenser fan cycling on pressure or temperature, a fan speed controller, or a flooding valve that backs liquid into the condenser to reduce its active surface. The minimum condensing pressure it is set to hold is whatever keeps enough differential across the metering device for that refrigerant and that evaporator temperature, so it is taken from the system design data rather than from a rule of thumb. Lowering the low-pressure cut-out only hides the symptom: the evaporator stays starved, refrigerant velocity stays too low to return oil, and the compressor runs short of both cooling and lubrication. A restricted drier is a year-round restriction, not a seasonal one, and it shows up as a temperature drop across the drier with high superheat; an overcharge raises head pressure rather than lowering it.

Q32. Answer: C

A phase monitor relay watches the three-phase supply and drops the control circuit when that supply is not fit to run a motor: a lost phase, two phases swapped, or phases too far out of balance. Phase loss is the worst of the three. A three-phase motor needs all three phases to make a rotating field, so with one phase open it cannot start from rest at all: it sits and hums and draws locked-rotor current through the two live phases. If it was already running when the phase opened it keeps turning on a pulsating single-phase field at greatly reduced capability, while current in the two remaining phases climbs and the windings heat quickly. Single-phasing is the leading cause of burnout on three-phase motors.

Phase reversal turns the motor backwards. Refrigerant flow does not reverse, it stops. A scroll compressor compresses in one direction only, so a scroll run backwards is very loud, delivers no cooling and is damaged within minutes; a screw compressor is direction-sensitive in the same way. A reciprocating compressor is far less sensitive, because its pistons reciprocate whichever way the crankshaft turns, but anything else fed from the same supply, condenser fans and pumps included, also runs backwards and loses its output. Phase imbalance is the quiet one: unequal phase voltages produce a considerably larger imbalance in the currents, and that extra current shows up as winding heat and shortened motor life.

The relay is a protective device, not a power supply. It cannot make a third phase out of a single-phase service, it does not redistribute load between the phases to save energy, and it senses nothing on the refrigerant side, so it has no way to know whether a system is undercharged. Phase protection is commonly specified on three-phase commercial equipment, and manufacturers normally call for it on direction-sensitive scroll and screw compressors precisely because reversed rotation ruins them. Confirm rotation after any supply work upstream of the unit.

Q33. Answer: B

Metering device (TXV or fixed orifice): creates the pressure drop separating the high and low sides of the system. The TXV (Thermostatic Expansion Valve) or fixed orifice/cap tube meters refrigerant flow, dropping pressure rapidly. This pressure drop causes partial flashing (evaporation), dramatically lowering refrigerant temperature. The resulting cold liquid/vapor mixture then enters the evaporator where it absorbs heat and fully evaporates.

Q34. Answer: C

Three distinct terms: Recovery (remove to cylinder), Recycling (on-site oil/moisture cleanup), Reclaiming (certified facility to AHRI 700 virgin purity). Recovery: extracting refrigerant from a system into an approved recovery cylinder — mandatory before opening. Recycling: passing recovered refrigerant through filters and oil separators for reuse on the SAME system or same owner's equipment. Reclaiming: sending to a certified reclaim facility for processing to AHRI Standard 700 purity (essentially virgin quality, sold for any use).

Q35. Answer: B

Indoor running but outdoor unit not starting: check power supply to the outdoor unit first. The indoor blower fan and outdoor compressor/condenser fan run on separate circuits. A tripped circuit breaker, open disconnect switch, or blown fuse at the outdoor unit disconnect kills the outdoor circuit while leaving indoor running. After confirming power: check the contactor (coil energized? contacts closed?), capacitor (start/run capacitors fail frequently — multimeter capacitance test), and check for safety control lockout (high/low pressure switch).

Q36. Answer: C

Condenser outlet: subcooled liquid under normal operation. The condenser removes heat from the refrigerant in three stages: 1) desuperheating (vapour cools to saturation), 2) condensing (vapour → liquid at constant pressure and temperature), 3) subcooling (liquid cools further, below saturation). Subcooling is what stops vapour bubbles forming in the liquid line before the metering device. Typical subcooling is 10–15°F (about 6–8°C), but the value to work to is the one on the equipment nameplate, since some equipment specifies outside that band.

Q37. Answer: D

Liquid refrigerant expands strongly with temperature and is very nearly incompressible, so a liquid-full cylinder has nothing left to absorb that expansion. Leave a vapour space and a warm afternoon simply walks the pressure up the saturation curve, which is what the cylinder is built for. Fill it solid and the same temperature rise has nowhere to go: pressure climbs hydrostatically, far faster and far higher than any saturation pressure, and the cylinder can rupture. That is why recovery cylinders are filled by weight, checked against the tare weight and the capacity marked on the cylinder itself, and why many are fitted with a float or a shut-off that stops the machine before the cylinder fills. Two related rules come from the same physics: never put a flame or other heat source on a cylinder to speed up a transfer, and never refill a disposable single-use cylinder, which has no overfill protection and was not built to be filled again.

Q38. Answer: D

A motor that hums at locked-rotor current has power and continuity but no starting torque, so look at the components that produce that torque. Voltage confirmed at the compressor terminals means the supply, disconnect and contactor have all done their job, which puts the fault in the motor or in what starts it. On a capacitor-start hermetic the starting torque comes from the start winding together with the start capacitor and the start or potential relay that switches it in and back out again; a PSC compressor has no start capacitor and no start relay at all, and on that machine the equivalent check is the run capacitor. A failed start capacitor, a relay whose contacts have burned open, or an open start winding leaves the rotor standing still while the run winding draws locked-rotor amps until the overload opens. Those parts are external, inexpensive and a common failure, so test them before condemning the compressor. Keep in mind that a mechanically seized compressor produces the same symptom, so if the start components test good the compressor itself becomes the suspect. Discharge any capacitor through a proper resistor before handling it.

Q39. Answer: D

R-22 (HCFC) = ozone depleter. Because of its ozone-depleting potential, Canada phased out R-22 production and import under the Montreal Protocol. As of 2020, R-22 can only be used from recovered/recycled stock. Common replacements include R-410A (new systems) and R-407C (retrofit).

Q40. Answer: D

LPC tripping = suction pressure falling below cut-out. Low refrigerant charge reduces suction pressure below the cut-out setpoint. Restricted liquid line, dirty evaporator coil, or low evaporator load can also cause low suction. The LPC protects the system from operating at pressures where evaporator freeze-up or compressor damage could occur.

Q41. Answer: A

After a compressor burnout, acid, carbon, and moisture are spread through the whole system and all of it must be removed before the replacement compressor runs. Acid formed by the thermal breakdown of oil and refrigerant attacks the winding insulation and plates copper onto bearing and valve surfaces, so a system left dirty destroys the new compressor. Procedure: 1) Recover the refrigerant. 2) Clean the lines and coils — flush with an approved flushing agent or blow through with dry nitrogen to carry out acid and carbon. 3) Fit an oversized burnout filter drier in the liquid line and a suction line drier ahead of the compressor. 4) Install the new compressor and charge it with the oil specified for that machine. 5) Triple evacuate — pull a vacuum, break it with dry nitrogen, repeat, then pull the final vacuum and prove it with a decay test, watching that the reading holds steady instead of creeping back up. 6) Charge and run the system, then acid-test the oil after the first day or two of operation; if acid or discolouration remains, change the drier cores and retest, and check the system again about two weeks later.

Q42. Answer: B

Boiling absorbs latent heat, and that is far more heat per kilogram than simply warming the refrigerant. Heat that turns a liquid into a vapour is latent heat, and for a refrigerant it is many times the sensible heat picked up by warming that same mass a few degrees. The evaporator is therefore built to boil the refrigerant off, and only the last stretch of the coil adds sensible heat as superheat. That is why a modest mass flow can carry a large cooling capacity, and why flooding or starving the coil, which changes how much of the coil is actually boiling, costs capacity so quickly. It is also why the refrigerant leaving the coil must be all vapour: any liquid still in it represents latent heat the coil failed to use, and it is heading for the compressor.

Q43. Answer: C

Floating head pressure: condenser fan speed or staging is controlled to allow condensing pressure to drop to the lowest stable level based on ambient conditions — improving system efficiency. Traditional systems maintain high fixed head pressure year-round. Floating head control reduces fans (or varies their speed) to allow the head pressure to drop as ambient drops in winter, reducing compressor work (lower pressure ratio = less energy per unit of refrigeration). The float is not unlimited, though: every floating head strategy is given a floor, because the metering device still has to be left with enough pressure difference across it to feed the evaporator, and refrigerant velocity has to stay high enough to carry oil back. The floor is worked out from the system design data for that refrigerant and evaporator temperature, and below it the same head pressure control described elsewhere in this bank takes over and holds condensing pressure up.

Q44. Answer: D

A relief discharge goes outside, not into the building. CSA B52 requires the pressure-relief device on a refrigerant pressure vessel such as a liquid receiver to discharge to the outdoors, terminating clear of doors, windows and fresh-air intakes so a release cannot re-enter the building or an occupied space. The vent line must be sized and routed so it cannot throttle the device's rated relieving capacity, and it must not be trapped, capped or valved off. Venting into the machinery room is wrong: machinery-room vapour detection and mechanical ventilation are separate requirements that deal with leaks, and they are never a substitute for a relief vent. Piping the relief into the suction line gives no protection at all, because it discharges into the same system it is supposed to protect. Leaving the outlet bare dumps the charge into the plant room. Confirm the termination heights and clearances in the edition of B52 adopted by your province.

Q45. Answer: C

Check cause before replacing compressor. Refrigerant flood-back causes liquid slugging that hydraulically locks compressor pistons. Low voltage at the compressor terminals causes high current + thermal overload. Compressor may free itself after cooling — but replacing without fixing the cause means the replacement fails too. Check: liquid slugging (suction line frost/wet), power supply and voltage at the terminals, overload conditions.

Q46. Answer: B

A cylinder that will hold recovered refrigerant is a pressure receptacle for the transport of dangerous goods, so it has to carry a specification Transport Canada authorizes and it has to be within its requalification date.

Transport Canada's own container pages describe the permanent TC markings a specification cylinder carries under CSA B339, and the requalifier's registered mark and date stamped on the cylinder each time it is requalified; CSA B340 governs the selection and use of those containers. Cylinders sold here are frequently dual-marked, because Transport Canada also lists DOT cylinders manufactured to the American rules among the authorized container types — so 'DOT-rated' on its own is US shorthand, and what actually matters is a specification Transport Canada accepts, plus a current requalification date.

A single-use disposable cylinder is out on both counts: it was never built to be refilled, it has no overfill protection, and refilling it is prohibited. A shop-built vessel carries no specification marking at all, whatever fittings are hung on it. And a pressure rating is not a specification: a cylinder can be strong enough for the gas and still be unmarked, unauthorized or long out of date. Fill by weight against the cylinder's tare and rated capacity so a vapour space is left, and keep a separate, clearly labelled cylinder for a burnout or for mixed refrigerant.

Q47. Answer: C

Voltage only tells the truth under load, and a drop measured between two points is made in the length of circuit between them. Standing still, a corroded lug, a loose terminal or a burnt contactor contact carries no current and drops no voltage, so everything measures perfectly. Ask the compressor to start and it draws locked-rotor current for a moment; that current through a high-resistance joint produces a large voltage drop, and what reaches the motor is no longer enough to develop starting torque. The rotor stays put, locked-rotor current continues, and the overload does its job. Because the two readings were taken at the same instant, they localize the fault: full voltage at the disconnect with low voltage at the terminals puts the loss between them - pitted contactor contacts, a loose or corroded connection, an undersized or badly spliced conductor. Look for discolouration and heat damage, and measure the drop across each connection and across the closed contactor while the machine is trying to start.

If the supply itself were weak, the disconnect reading would sag along with the terminal reading instead of holding at 240 V. A failed start capacitor leaves full voltage standing at the terminals while the motor hums, so it puts no drop between the two measuring points. A seized compressor also draws locked-rotor current, but through a sound supply path the terminal voltage would stay close to the disconnect reading rather than falling 50 volts below it.

Q48. Answer: D

All refrigerant must be recovered with certified recovery equipment into an approved recovery cylinder. In Canada that means a TC specification cylinder. Transport Canada's requirements for manufacturing, inspection, testing, marking and requalification of cylinders are set out in CSA B339, which CSA B340 invokes for TC specification cylinders authorized for use in Canada under the Transportation of Dangerous Goods Regulations. DOT is the United States Department of Transportation, so a DOT stamp is the American specification; cylinders sold in North America are commonly marked to both, and the marking to look for on this side of the border is TC. Releasing refrigerant to atmosphere is prohibited: the Federal Halocarbon Regulations, 2022 made under CEPA 1999 forbid it for systems under federal jurisdiction - Crown-owned systems, federal works and undertakings, and systems on federal or Indigenous lands - and everywhere else the use and handling of these substances are regulated by the province or territory through its own halocarbon regulation. There is no small-quantity exemption and no acceptable way to vent slowly. Never mix refrigerants in one cylinder; a mixed cylinder cannot be reclaimed and becomes waste for destruction.

Q49. Answer: A

Blocked condensate drain (algae, mold, debris) is the most common cause of indoor water leakage — water backs up, the pan overflows, and it spills from the unit. The condensate drain pan collects water that drips off the evaporator coil. A blocked drain causes the pan to fill and overflow. Algae growth in the drain line (common in humid environments) is the most common blockage cause. Additional causes: incorrect drain pan slope (should slope toward drain), cracked/leaking drain pan, drain line installed without adequate slope, condensate pump failure.

Q50. Answer: C

A coil that comes out of defrost completely clear has had enough heat; if the cycle still ran to its time limit, the control never saw the coil warm up. Temperature termination is how an electric defrost is meant to end - once the ice is gone the coil metal climbs quickly past the termination setting and the cycle stops, which saves energy and stops the box temperature climbing any further. The 30-minute limit is a fail-safe, not the intended terminator. Here the ice is gone and the drain pan is hot, so heat input and frost load are not the problem; what has failed is the device that is supposed to notice. Either the termination thermostat has failed open or drifted out of calibration, or it is clamped where it never sees warm metal - on a return bend that stays cold, or away from the heater path. Test it for continuity as its temperature rises and confirm it is mounted where the equipment's installation instructions place it. Heaters that were undersized for the coil and its load would leave ice still on the coil when the timer ended, which is exactly what is not found here. An undercharge is a refrigeration-side fault and has no bearing on whether a defrost cycle terminates. And a defrost is not supposed to run a fixed length regardless of need - preventing that is precisely why temperature termination exists.

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